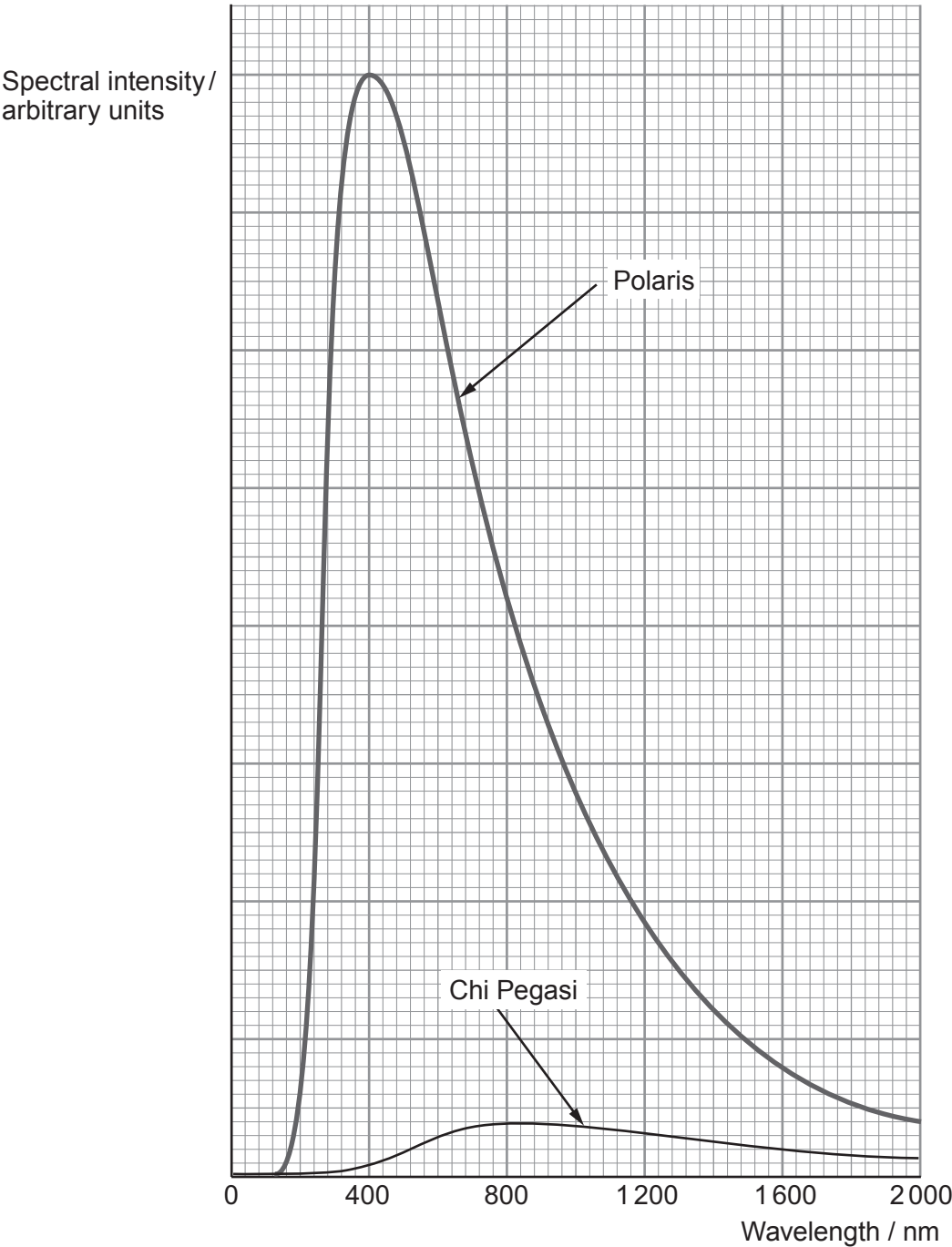


4. (a) Stars are very good approximations to *black bodies*. State what is meant by a *black body*. [1]

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(b) The graph shows the black body radiation curves for the two stars Polaris (sometimes called the North Star) and Chi Pegasi (a red supergiant in the constellation Pegasus). The stars are equidistant from the Earth.



Examiner
only

(i) Use the graph to state **three** differences between Polaris and Chi Pegasi. [3]

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(ii) The surface temperature of Polaris is 7 250 K. How does the graph confirm this? [2]

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(iii) Polaris is 431 light years from Earth and the intensity of radiation received on Earth from it is $4.05 \times 10^{-9} \text{ W m}^{-2}$. Show that the luminosity of Polaris is approximately $8.5 \times 10^{29} \text{ W}$. [1 light year = $9.46 \times 10^{15} \text{ m}$] [2]

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(iv) Calculate the radius of Polaris. [3]

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